JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHIVAM AGENCIES IOC DEALERS-KSK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

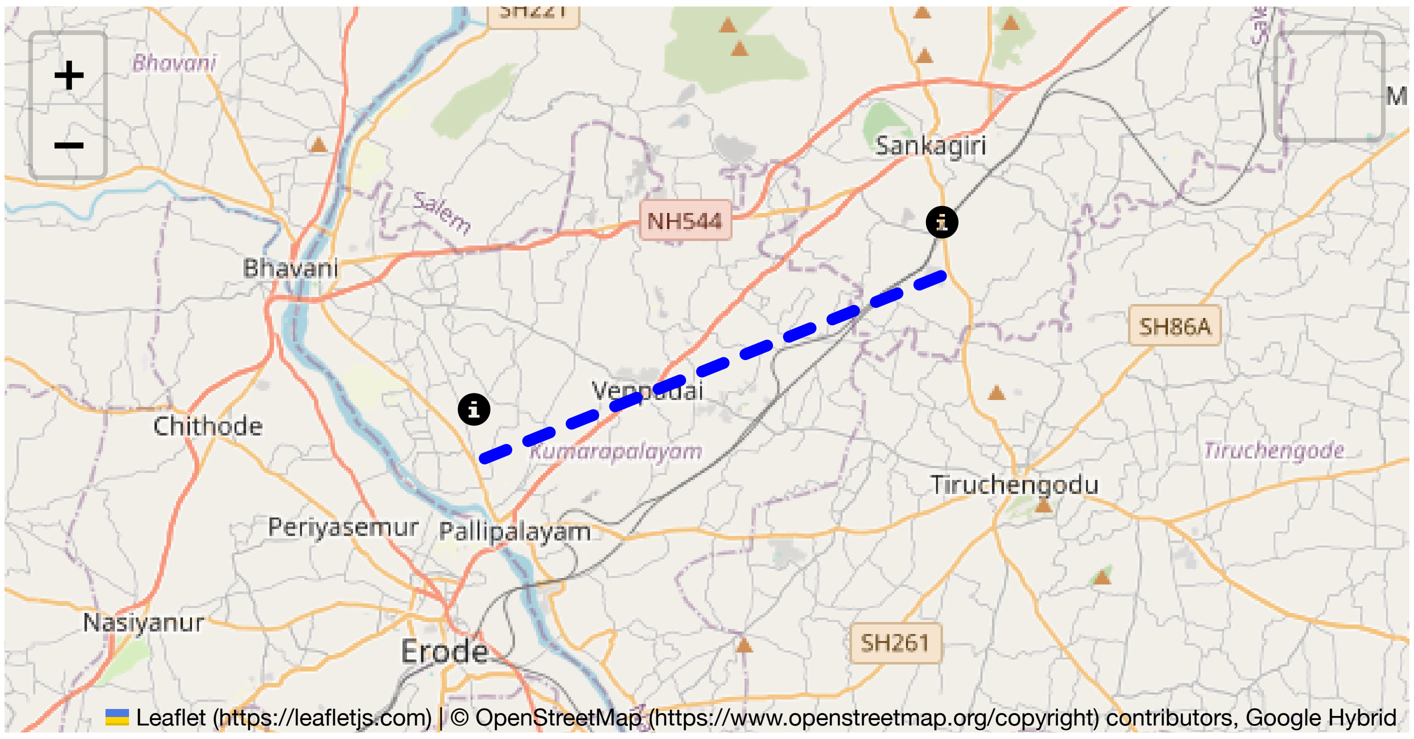
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

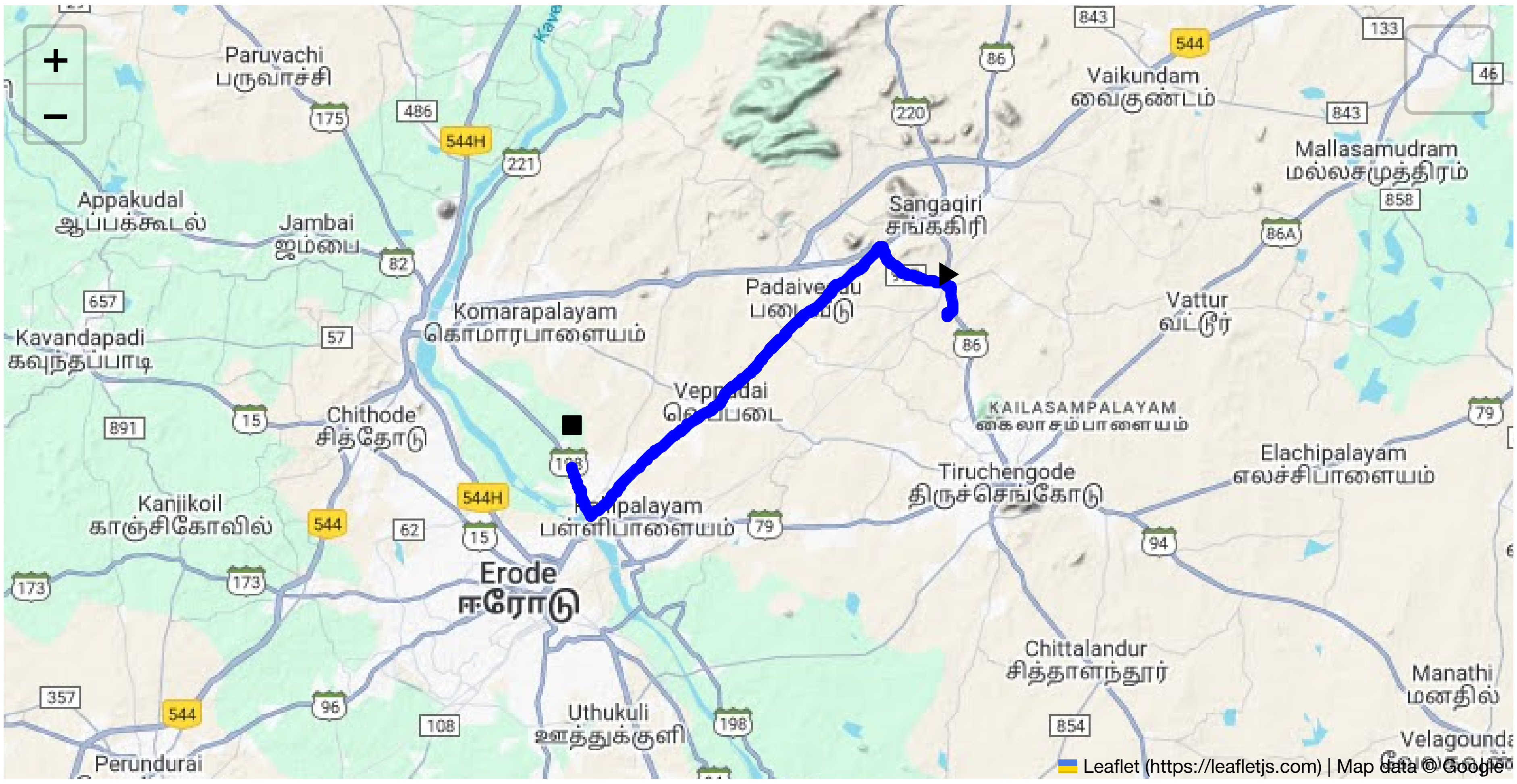
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 22.79 km
Estimated Duration: 0.5 hours

Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.384551, 77.736069)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 9PMP+V9H, Ganapathipalayam, spans approximately 22.79 kilometers. This route is typically traveled within 0.55 hours by trucks carrying hazardous materials. It primarily involves local roads and highway segments vital for transit in Tamil Nadu.

Weather Conditions

Typical Weather Conditions:

- **Summer (March to June):** Temperatures can exceed 40°C, contributing to asphalt softening and potential road surface degradation.
- **Monsoon (July to September):** Heavy rains can lead to flooding, especially in low-lying areas, reducing visibility and road traction.
- **Winter (October to February):** Mild and cooler temperatures with occasional fog in early mornings.

Weather-Related Hazards:

- **Monsoon floods** might impact travel times due to waterlogged roads.
- **Heat waves** could affect vehicle performance and increase the risk of tire blowouts.

Traffic Patterns

Peak Hours and Congestion:

- **Morning (8:00 AM to 10:00 AM) and Evening (5:00 PM to 7:00 PM):** Local roads experience increased congestion due to daily commutes.
- Congestion-prone areas include market zones and major intersections.

Road Quality and Infrastructure

- **Main Roads:** Generally well-maintained but can suffer from wear during monsoon seasons.
- **Secondary Roads:** Some segments may have potholes or unpaved sections requiring careful navigation.

Alternative Routes

- **Emergency Detours:** Consider local roads that parallel main highways for detours in case of accidents or roadblocks. State Highway 86 can be used as an alternate.

Local Regulations

- **Hazardous Material Transport:** Requires adherence to Tamil Nadu's regulations, including proper documentation and marked signage on vehicles.
- **Time Restrictions:** There might be time restrictions on heavy vehicles entering certain areas, especially during peak traffic hours.

Historical Incidents

- There have been occasional reports of accidents involving trucks primarily due to wet road conditions or vehicle malfunctions. No major hazardous material incidents reported recently.

Environmental Considerations

- **Sensitive Areas:** Parts of the route pass through agricultural land, and caution is advised to prevent spills that could contaminate local ecosystems.

Communication Coverage

- **Cellular Coverage:** Most of the route is covered by mobile networks; however, rural stretches might experience brief dead zones. It is advisable to check network maps prior to travel.

Emergency Response Times

- **Urban Segments:** Response times are relatively quicker, averaging around 15-20 minutes.
- **Rural Segments:** May face delays, taking 30 minutes or more due to distance from emergency services.

Overall Risk Assessment Summary

- **Key Risks:** Flooding during the monsoon, traffic congestion during peak hours, and tire or vehicle issues in extreme heat.
- **Mitigation Measures:** Proper weather awareness, maintaining safe driving practices, compliance with transport regulations, and emergency readiness are crucial.
- **Recommendation:** Regular maintenance checks on vehicles and real-time traffic and weather monitoring to adapt promptly to changing conditions.

This route, while generally manageable, requires attention to traffic patterns and weather conditions to ensure safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.37086, 77.74828	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.46307, 77.84941	30 KM/Hr
6	Turn	Medium	11.46318, 77.84913	30 KM/Hr
7	Turn	Medium	11.46312, 77.84886	30 KM/Hr
8	Blind Spot	Blind Spot	11.36726, 77.74328	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Cheran Hospital	11.3680302, 77.7481976	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Ssri Valliappa Vidhayalayam mat hr school	11.4004878, 77.7794381	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37086, 77.74828

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46312, 77.84886

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.36726, 77.74328